Modelling the energy levels of atoms using the Hartree Fock method on a radial finite difference grid.

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Abstract

The Hartree–Fock method provides an approximate approach for determining the energy levels and total energy of atomic systems. It employs a combination of theoretical formalism and numerical techniques to address cases where analytical solutions are not available. This report presents a study on firstly Lithium following onto Beryllium, Boron and Carbon's energy levels using the Fock matrix method. It is intended that theory and the accompanying code may serve as a resource for future students, helping to bridge the gap between the theoretical framework and practical implementation of approximate quantum methods.

1 Introduction

The Schrödinger equation for hydrogenic systems is typically introduced during the early stages of undergraduate study, where it can be solved analytically to obtain exact energy levels. However, for multi-electron atoms the problem becomes significantly more complex. For example with helium, although approximate treatments can reduce the system to effective two-body problems, an exact analytical solution is no longer attainable, necessitating the use of numerical methods. As the number of electrons increases, the complexity of the governing equations grows rapidly due to electron–electron interactions. In systems containing more than two electrons, additional exchange terms arise for electrons with parallel spin, reflecting the antisymmetry requirement of the total wavefunction imposed by the Pauli exclusion principle. This leads to observable effects on electron binding and energy levels. The aim of this project was to introduce a Hartree Fock method by following the textbook by Szabo and Ostlund [1], and to provide a bridge between this and the steps to reproducing a numerical method. First the theory and how we approach these problems must be covered. Then the report will discuss how to code the problem and the differences between the closed and open shells resulting in the restricted and un-restricted model. And then the different procedure for Boron as we diverge from a spherical approximation. The results and accuracy of the code and finally the uses of the energy levels will be discussed. The atoms this report covers is that of Lithium, Beryllium, Boron and Carbon by extension it should be possible to model all atoms if the orbital shapes are modeled as spheres which is what is done in Boron and Carbon, these examples will be discussed.

2 Theory

2.1 Many body problem and Hamiltonian

This project looks to solve the energy levels of atoms larger than helium, we will be working in atomic units, the Hamiltonian that is the focus of the study is such:

$$\hat{H} = -\sum_{i=1}^{N} \frac{1}{2}\nabla_i^2 - \sum_{i=1}^{N} \frac{Z}{r_i} + \sum_{i=1}^{N}\sum_{j>i}^{N} \frac{1}{r_{ij}} \tag{1}$$

This operator consists of three main components: the kinetic energy of each electron, the Coulomb attraction between the positively charged nucleus and each electron, and finally, the electron-electron interaction. The last term is what introduces the primary challenge of the many-body problem, as the $1/r_{ij}$ term intricately couples the motions of all electrons together.

To find the exact energy levels of the atom, we must solve the time independent Schrödinger equation.

$$\hat{H}\Psi = E\Psi \tag{2}$$

Because the electron-electron repulsion term makes this equation analytically unsolvable for systems with more than one electron, we must rely on advanced approximation methods. The model we will used in the central finite difference approximation, and we reduce the wavefunction into separate spherical and radial components. It is possible to solve just the radial equation due to spherical symmetry. This can be done through the central field approximation which is an assumption that every electron moves independently in an average potential. Meaning each electron sees a central potential that is only a function of r (radial distance). This is possible as the radial and spherical components can be separated to satisfy there own equations [2]. This is not to say we only deal with spherical potentials, when we move from the s orbitals to the first p orbits as in Boron the electron shells are a dipole shape, we deal with this with a slightly different term in the multi-pole expansion. This allows us to reduce our Eq.1 to [3]:

*Ignoring the electron-electron term for now.

$$\left[-\frac{1}{2}\frac{d^2}{dr^2} + V(r) + \frac{l(l+1)}{2r^2}\right]u(r) = \epsilon u(r) \quad (3)$$

Now we are able to use a radial grid to solve this equation. In this expression, the kinetic energy operator is simplified to a second-order derivative with respect to r, which can be mapped directly onto a finite difference grid as seen with Eq.4:

$$\frac{d^2u}{dr^2} = \frac{u_{i-1} - 2u_i + u_{i+1}}{r^2} \quad (4)$$

This has a simple matrix form of -2 along the diagonal and +1 on either side - a triangular matrix [4]. When solving these problems computationally we transition from continuous wavefunctions to a discrete set of points, following the Roothaan-Hall approach as outlined by Baskerville [5], allowing for the use of standard linear algebra eigensolvers. This is represented by our radial grid a vector, which we can write as a matrix being necessary later. The second term in Eq.3, V(r) represents the attractive nuclear potential,$-Z/r$, and the final term, $\frac{l(l+1)}{2r^2}$, represents the centrifugal term. This is the angular momentum representation, this being dependant on l only needs to be considered for larger atoms such as Boron and Carbon for our study. It introduces a repulsive barrier near the nucleus. This physically prevents electrons with higher angular momentum from penetrating the core as effectively as s-electrons.

By defining the Hamiltonian in terms of the reduced radial wavefunction u(r), the numerical task becomes the diagonalization of a triangular matrix where the diagonal elements contain the potential and centrifugal terms at each grid point r_{ij}. This approach streamlines the computation and shows the $1/r^2$ dependence of the centrifugal term underscores the necessity of a high point density near the origin. The electron-electron interaction term is a bit more complicated than the above so it is essential that more theory is covered before coming to the terms which represent this. Once this is done we can combine our matrices to solve for the energy levels of the system.

2.2 The wavefunction and the slater determinants

The last term from Eq.1 needs to be represented. The wavefunction we are interested in must obey the Pauli exclusion principle which can be seen in the results of the energies of the different electrons. Electrons with the same spin can not occupy the same area of space so on average the electrons will be farther from each other which can reduce or increase the energy of the electron depending on the case, this has to be considered alongside the 'standard' electron-electron coulomb repulsion. As we would expect there is a repulsion term between the two electrons we look at for a given time. There is also going to be an additional second term to consider due to Pauli's exclusion principle. The anti-symmetry requirement of the multi-electron wavefunction is satisfied by the use of a Slater Determinant. [6], an example is shown below in figure 1. This would be the case for Lithium, therefore we build a 3x3 matrix as there are 3 electrons in the system, we must consider both the spin and position.

* Make here a table for your own Lithium slater determinant so don't have to reference.

The determinant of the matrix can be calculated and there are 6 terms that survive, the terms that have 0 or two variable changes remain as they are and the terms with one variable change flip sign and become negative this is because the wavefunction of electrons must be completely antisymmetric with respect to the exchange of any two particles. Mathematically, swapping the coordinates of two electrons is the exact same as multiplying the entire determinant by -1. The slater determinant is used to build the backbone of the code and is integrated into the problem itself rather than using an actual slater determinant as the initial guess. Meaning we have to consider two types of interaction which we will call the coulomb and the exchange terms which we will be able to understand shortly and the calculation can be explained.

2.3 Deriving the Hartree Fock equations using the variational principle and langrange multipliers

The derivation for the Hartree Fock equations will be shown now and follow close to [1] method to ensure a good understanding of the problem. First we will use the variational principle to minimise the energy expression for a slater determinant, later we shall look at the difference when introducing assumptions around spin and how this will allow us to solve the equations we are building. The variational principle allows us to start with a trial function and minimise this point, it will always be greater than that of the actual value we are interested in but will give a close value. To approximate the electronic wave function Φ, we expand it as a linear combination of a set of N known basis functions $\{\Psi_j\}$:

$$\Phi = \sum_{j=1}^{N} c_j \Psi_j \tag{5}$$

Our goal is to minimize the energy expectation value $E = \langle \Phi | \mathcal{H} | \Phi \rangle$ subject to the normalization constraint $\langle \Phi | \Phi \rangle = 1$. Using the method of Lagrange multipliers, we define the functional $\mathcal{L}$:

$$\mathcal{L} = \langle \Phi | H | \Phi \rangle - E(\langle \Phi | \Phi \rangle - 1) \tag{6}$$

Substituting our linear expansion into the functional gives:

$$\mathcal{L} = \sum_{i,j} c_i^* c_j \langle \Psi_i | H | \Psi_j \rangle - E \left(\sum_{i,j} c_i^* c_j \langle \Psi_i | \Psi_j \rangle - 1 \right) \tag{7}$$

By defining the Hamiltonian matrix elements $H_{ij} = \langle \Psi_i | \mathcal{H} | \Psi_j \rangle$ and the overlap matrix elements $S_{ij} = \langle \Psi_i | \Psi_j \rangle$, we require the first variation of $\mathcal{L}$ with respect to the coefficients c_i^* to vanish as $\delta_{c_i^*} \mathcal{L} = 0$. This leaves us with the set of equations:

$$\sum_{j=1}^{N} (H_{ij} - E S_{ij}) c_j = 0 \tag{8}$$

In matrix form, this results in the generalized eigenvalue problem:

$$\mathbf{Hc} = E\mathbf{Sc} \tag{9}$$

Now moving onto spin orbitals we minimise the energy again but this time with a different constraint, the spin orbitals,

To determine the optimal spin orbitals $\{\chi_a\}$, we minimize the electronic energy E_0 subject to the orthonormality constraint $\int dx_1 \chi_a^*(1) \chi_b(1) = \delta_{ab}$. We introduce a Lagrangian $\mathcal{L}$ with a matrix of Lagrange multipliers ϵ_{ba}:

$$\begin{aligned} &\mathcal{L}[\{\chi_a\}] = E_0[\{\chi_a\}] \\ &- \sum_{a=1}^{N} \sum_{b=1}^{N} \epsilon_{ba} \left(\int dx_1 \chi_a^*(1) \chi_b(1) - \delta_{ab} \right) \end{aligned} \tag{10}$$

where E_0 is the expectation value of the single Slater determinant. This links directly to our Hamiltonian and what we will be solving for numerically, the first term is the kinetic and nucleus attraction, the second term is the coulomb term and the exchange term in the summation:

$$E_0 = \sum_{a=1}^{N} \langle a | h | a \rangle + \frac{1}{2} \sum_{a,b}^{N} ([aa|bb] - [ab|ba]) \tag{11}$$

Requiring the first variation of $\mathcal{L}$ to vanish ($\delta \mathcal{L} = 0$) leads to the Hartree-Fock integro-differential equations, notice we are left with coupled equations:

$$f(1) \chi_a(1) = \sum_{b=1}^{N} \epsilon_{ba} \chi_b(1) \tag{12}$$

Here, $f(1)$ is the Fock operator, defined as the sum of the core-Hamiltonian $h(1)$ and the effective potential generated by all occupied orbitals:

$$f(1) = h(1) + \sum_{b=1}^{N} (J_b(1) - K_b(1)) \tag{13}$$

The Coulomb operator J_b and Exchange operator K_b are defined by their actions on an arbitrary orbital $\chi_a(1)$:

$$J_b(1) \chi_a(1) = \left[\int dx_2 \chi_b^*(2) \frac{1}{r_{12}} \chi_b(2) \right] \chi_a(1) \tag{14}$$

$$K_b(1) \chi_a(1) = \left[\int dx_2 \chi_b^*(2) \frac{1}{r_{12}} \chi_a(2) \right] \chi_b(1) \tag{15}$$

By performing a unitary transformation on the occupied orbitals, we diagonalize the matrix ϵ_{ba}, reducing the coupled equations to the canonical form:

$$f(1) \chi_a(1) = \epsilon_a \chi_a(1) \tag{16}$$

This equation or set of equations will reappear constantly and will be the focus of our study, we will aim to solve the equations to obtain energy values for our electron orbitals and use these to calculate the energy of the atom to compare with other studies and experimental results.

When calculating the total energies it would be easy to assume that this is just equal to the sum of all of the orbitals, however it is important to note that this would result in the double counting of all of the electron electron interactions between pairs, so we must account for this. The total energy of the atom can be calculated in Eq11. This can be simplified. The sum of the individual orbitals is as seen in Eq. where the electron - electron interaction is double counted it is possible to reach the final energy with the addition of the core energy values and multiplying this sum by 0.5. This average solution is neat and easy.

2.4 The density matrix + the coulomb and exchange operators

To calculate the Coulomb ($\hat{J}$) and Exchange ($\hat{K}$) operators for our Fock matrix, we first need a mathematical map of the electron distribution. The probability of finding an electron in a specific region of space is given by the total electron density, $\rho(\mathbf{r})$, which is the sum of the squared modulus of the occupied molecular wavefunctions.

For a closed-shell system with N electrons (where each spatial orbital is doubly occupied), we sum over the $N/2$ occupied spatial orbitals. By expanding these unknown molecular orbitals (ψ_a) into our known basis functions (ϕ_μ), we can extract the density matrix, $P_{\mu\nu}$:

$$\begin{aligned}
\rho(\mathbf{r}) &= 2\sum_a^{N/2} \psi_a^*(\mathbf{r})\psi_a(\mathbf{r}) \\
&= 2\sum_a^{N/2}\sum_\nu C_{\nu a}^*\phi_\nu^*(\mathbf{r})\sum_\mu C_{\mu a}\phi_\mu(\mathbf{r}) \\
&= \sum_{\mu\nu}\left[2\sum_a^{N/2} C_{\mu a}C_{\nu a}^*\right]\phi_\mu(\mathbf{r})\phi_\nu^*(\mathbf{r}) \qquad (17)\\
&= \sum_{\mu\nu} P_{\mu\nu}\phi_\mu(\mathbf{r})\phi_\nu^*(\mathbf{r})
\end{aligned}$$

The density matrix, $P_{\mu\nu}$, is the discrete, computational representation of the electron density. It is the crucial link in our iterative process. In the Hartree-Fock approximation, a single electron is modeled as feeling the repulsion of all other electrons not as individual moving point charges, but as an average, static field. However, to calculate this average field (the potential), we need to know the wavefunctions of the electrons. Yet, to find the wavefunctions, we must first solve the Schrödinger equation, which requires the potential. This paradox is resolved computationally via the Self-Consistent Field (SCF) method. Through the use of initial guesses for the wavefunctions, we can build an initial density matrix. This matrix is then used to construct the $\hat{J}$ and $\hat{K}$ operators, which form the Fock matrix to be diagonalized, yielding a new, improved density matrix. This is one of the main approximations we make as we should have more variables than we do at a given time to calculate our energy but through iteration we can obtain accurate solutions nonetheless.

With the density matrix defined, the next computational challenge is constructing the specific expressions for the Coulomb ($\hat{J}$) and Exchange ($\hat{K}$) operators. Transitioning from continuous equations to numerical approximations significantly simplifies the problem: our continuous spatial wavefunctions are evaluated on a 1D grid to become vectors, meaning our quantum mechanical operators can be represented as matrices.

The Coulomb operator represents the classical electrostatic repulsion. The potential created by an electron in orbital j acting on an electron in orbital i is defined as:

$$\hat{J}_j(r_1)\psi_i(r_1) = \left[\int \frac{|\psi_j(r_2)|^2}{r_{12}}dr_2\right]\psi_i(r_1) \qquad (18)$$

By summing over all occupied orbitals, the total Coulomb potential simply relies on the total electron density (ρ). Because this is a local operator—the potential at r_1 only depends on the surrounding charge distribution—computationally, it evaluates to a 1D vector. To incorporate this into the Fock matrix, this vector is placed along the diagonal of the J matrix, with all off-diagonal elements set to zero.

The Exchange operator, however, is purely a quantum mechanical consequence of the Pauli exclusion principle making it more difficult. Based on the derivation by Szabo and Ostlund, its action on a wavefunction involves swapping the coordinates of the two electrons:

$$\hat{K}_j(r_1)\psi_i(r_1) = \left[\int \frac{\psi_j^*(r_2)\psi_i(r_2)}{r_{12}}dr_2\right]\psi_j(r_1) \qquad (19)$$

By factoring out $\psi_i(r_2)$ from the integral and summing over all occupied orbitals j, we can express the total exchange operator in terms of the non-local density matrix, $P(r_1, r_2) = \sum_j \psi_j(r_1)\psi_j^*(r_2)$:

$$\hat{K}(r_1, r_2)\psi_i(r_1) = \int\left[P(r_1, r_2)\frac{1}{r_{12}}\right]\psi_i(r_2)dr_2 \qquad (20)$$

Because the operator involves evaluating the wavefunction at two different coordinates simultaneously, it is non-local. In our computational program, this means the K matrix cannot be diagonalized into a single vector like J. Instead, it forms a full matrix where the elements are computed as the element-wise product of the density matrix and the $1/r_{12}$ interaction kernel.

The critical term linking both of these operator calculations is $1/r_{12}$. Read on to find out how this specific interaction is evaluated.

2.5 Multipole expansion theorum - shapes of the atomic orbitals

The Coulomb and Exchange operators fundamentally rely on evaluating the electron-electron repulsion term, $1/r_{12}$. To compute this within the framework of a 1D radial grid, this term must be decoupled into its radial and angular components using the Laplace multipole expansion [7]:

$$\frac{1}{r_{12}} = \sum_{l=0}^{\infty} \sum_{m=-l}^{+l} \frac{4\pi}{2l+1} \frac{(r_<)^l}{(r_>)^{l+1}} \times Y_{lm}^*(\theta_1, \phi_1) Y_{lm}(\theta_2, \phi_2) \quad (21)$$

For the first two atomic structures of interest, Lithium and Beryllium, the calculations only involve s-orbitals. Because the electron density cloud of an s-orbital is spherical, $l = 0$. This simplifies the mathematics significantly; the spherical harmonics equal a constant $1/\sqrt{4\pi}$, which cancels out the 4π in the expansion, removing all angular dependence. The radial terms reduce to $(r_<)^0/(r_>)^1$, leaving simply $1/r_>$, where $r_>$ represents the greater of the two radial coordinates being compared. Computationally, this allows the interaction kernel to be built in a single line using a maximum value function.

However, for higher values of l, such as $l = 1$ in the case of Boron's $2p$ electron, the system is no longer spherical. Expanding the $l = 1$ term explicitly yields the dipole interaction[8]:

$$\text{Dipole Term} = \frac{r_<}{r_>^2} \Big(\cos\theta_1 \cos\theta_2 + \sin\theta_1 \sin\theta_2 \cos(\phi_1 - \phi_2) \Big) \quad (22)$$

(This term is short than writing out the spherical harmonic but maybe it would be good to have these aswell.) The use of this angular dependance will in all cases be integrated through the calculation for K matrices (The J matrix is calculated with a spherical approximation). As nothing else depends on this result it is possible to calculate seperately the effect of the angular difference between a dipole and a sphere and just substitute the coefficient into the final equation.

The full term we have for the spherical harmonic dependence and sum over m=+-l without the radial component is :

$$\begin{aligned} \text{Angular dependence} = {} & Y_{1,-1}^*(\theta_1, \phi_1) Y_{1,-1}(\theta_2, \phi_2) \\ & + Y_{1,0}^*(\theta_1, \phi_1) Y_{1,0}(\theta_2, \phi_2) \\ & + Y_{1,1}^*(\theta_1, \phi_1) Y_{1,1}(\theta_2, \phi_2) \end{aligned} \quad (23)$$

This we need to integrate with respect to a sphere and dipole spherical harmonics choosing just one axis for the dipole we can see that the integral will cancel out to the result of: 1/3 with this formula:

$$\int Y_{L_1,m_1}(\theta, \phi) Y_{L_2,m_2}(\theta, \phi) = \delta_{l_1,l_2} \delta_{m_1,m_2} \quad (24)$$

We can perform this integral:

$$\frac{4\pi}{3} \int Y_{0,0}^*(\theta, \phi) Y_{1,0}(\theta, \phi) \times [\text{interaction(eq.7)}] \times Y_{0,0}(\theta, \phi) Y_{1,0}^* \, d\Omega \quad (25)$$

This integral produces a term to use for calculation of the exchange term between spherical and dipole shapes.

While the multipole expansion extends to infinity, calculating just the first non-vanishing term is sufficient for the ground state of Boron. Higher-order terms (like $l = 2$) would only become strictly necessary if we were calculating the interactions of d-orbitals in transition metals.

2.6 Restricted and unrestricted hartree Fock models - Koopmanns theorum

Having established the formalism for constructing the Fock matrix, we now distinguish between the restricted and unrestricted Hartree–Fock (HF) methods, and their applicability to closed- and open-shell systems.

In the restricted Hartree–Fock (RHF) method, each spatial orbital is doubly occupied by electrons of opposite spin. Consequently, the spin-up (α) and spin-down (β) electrons share the same spatial orbital, and the total electron density can be described using a single density matrix. This approach is therefore appropriate for closed-shell systems, such as Beryllium which was the first atom we looked at having 2 electron pairs in each shell.

For such systems, the Fock matrix is constructed as

$$F = T + V + J - K, \quad (26)$$

Starting from an initial guess for the orbitals , $1s$ and $2s$, one constructs the density matrix, from which the Coulomb and Exchange matrices are evaluated. Solving the resulting Fock matrix eigenvalue problem yields improved approximations to the orbital energies, with the lowest eigenvalues corresponding to the occupied orbitals.

Within this framework, Koopmans' theorem provides an approximate interpretation of orbital energies, stating that the ionisation energy is approximately equal to the negative of the corresponding orbital energy, assuming no relaxation of the remaining orbitals.

While RHF is efficient and well-suited to closed-shell atoms, it becomes inadequate for open-shell systems. In such cases, the unrestricted Hartree–Fock (UHF) method is preferred for example with Lithium. Here if one was to use the same method then an exchange term between an α spin and β spin electron would be calculated however this interaction does not exist. The UHF method by Pople-Nesbet from the core textbook [1] follows the same process for derivation as we have seen, but uses a combination of different density matrices to represent each spin and in some cases later each orbital shape. This leads to two coupled Fock equations with a similar derivation to earlier:

$$F^{\alpha} = T + V + J[P^{\alpha} + P^{\beta}] - K[P^{\alpha}], \quad (27)$$

$$F^{\beta} = T + V + J[P^{\alpha} + P^{\beta}] - K[P^{\beta}]. \quad (28)$$

This correctly reflects the physical fact that exchange interactions do not occur between electrons of opposite spin. The equations may be solved separately because the exchange interaction only occurs between electrons of the same spin, allowing the total Fock matrix to decouple into independent alpha and beta equations that share only the total electrostatic potential.

For atoms such as Lithium or Boron, which contain unpaired electrons, this results in separate equations for each spin set of electrons, allowing greater flexibility in the wavefunctions and improved accuracy compared to RHF. Furthermore, with Boron it requires a density matrix for the p orbital due to a different interaction coefficient with the shape of Boron's p orbitals no long being spherical.

For higher-order orbitals, such as those encountered beyond the ground-state of Beryllium, the assumption of spherical symmetry is no longer strictly valid. In particular, p-orbitals possess angular dependence. Since the present simulation framework relies on a spherically symmetric potential, an additional approximation is required.

To maintain spherical symmetry, the occupation of a single p-orbital (e.g., p_x) is replaced by an equal distribution over all three degenerate p-orbitals. That is, the electron density is averaged such that each of p_x, p_y, and p_z carries one-third of the occupation. This procedure effectively restores a spherically symmetric charge distribution, allowing the use of the central field approximation.

However, this averaging introduces additional contributions to the electron–electron interaction terms. In particular, when considering interactions between s and p orbitals, it becomes necessary to include dipole-like corrections arising from the non-spherical character of the underlying orbitals. These terms ensure that the interaction between the averaged p-electron density and the s-orbitals is treated consistently within the approximation.

2.7 Note on orthogonality and the spectral theory and how it just works

In the Hartree Fock model it is essential that the resulting orbitals form an orthonormal set in both the restricted and unrestricted case. In many standard quantum chemistry implementations, the use of non-orthogonal basis sets (such as Gaussian-type orbitals) leads to a generalised eigenvalue problem of the form [FC = SCE,] as we saw earlier in the derivation. where S is the overlap matrix. In such cases, an explicit orthogonalisation procedure is required. I had originally thought it could be an idea to use methods such as gramm-schmidt othoganalisation whether this would be mapping one solution to another or an alternation through the iteration was under consideration.

However we implement a finite difference grid, this corresponds to a discretised real-space basis which is inherently orthogonal, such that the overlap matrix reduces to the identity ($S = I$). As a result, the eigenvalue problem simplifies to [FC = CE,] and the resulting eigenfunctions are automatically orthonormal without the need for additional orthogonalisation procedures such as Gram–Schmidt. This behaviour can be understood in terms of spectral theory [9]: the discretised Fock operator is represented by a Hermitian matrix, whose eigenvectors form an orthogonal basis. Consequently, orthonormality is guaranteed by the mathematical structure of the problem rather than requiring explicit enforcement.

3 Methods and code

3.1 Initial set up and functions

Having established the underlying theory, we can now examine the practical implementation of the code and the specific roles of its components.

To start we must define some initial conditions, including the finite difference grid. This is an array of N points spaced by a small interval dr. To avoid numerical instabilities and the $1/r$ or $1/r^2$ singularity at the nucleus, the grid starts at dr rather than zero. Other essential parameters include the grid size this being tailored to the expected atomic orbital radius and the atomic number Z, which defines the number of protons for the atom of interest.

We must also provide the initial guesses for the wavefunction that will be updated iteratively, the reduced hydrogenic wavefunctions were first trialed, the aim is to have the solutions to converge quite quickly so the initial wavefunction choice is something that will be explored.

The aim now is to construct the terms in Eq.1 in a matrix form so it is possible construct a total matrix which we can use with the initial wavefunctions to solve for the next iteration of wavefunction and ultimately the energy levels.

Next the code for the functions, first for the kinetic energy term, we have the T matrix can be made simply as a combination of -2 on the diagonal and 1 on the off centre this is simple and can be seen in the appendix. The V matrix on the diagonal is also very simple as you would expect being equal to -Z/r.

The r_{12} multi-pole term is a bit more complicated to make, in the monopole instance this needs to take in again all values of r and create a grid in which the values are greater, originally I had an if statement for this but it can be achieved with the outer product function and taking the maximum value at each position. The dipole situation is slightly different as the value changes as per Eq.22 there there is a $r >$ and $r <$ term but following the same procedure we can calculate a grid for this as-well, these are kept as separate functions as it helped keep things simpler when working with these problems. The angular dependence we can keep separate and deal with in the K matrix.

Then we have the density matrices ($\mathbf{P}$), this function will take in the initial guesses of the wave functions and compute the density matrix seen in Eq.17, from here we can use the density matrices in the functions for the $\mathbf{J}$ and $\mathbf{K}$ matrices the last two to calculate. For the $\mathbf{J}$ matrix, we need the charge density which is the diagonal values of $\mathbf{P}$, this is taken and integrated alongside the value from $1/r_{12}$ kernal as per Eq.18. For the $\mathbf{K}$ matrix we follow Eq.19 which is the $\mathbf{P}$ matrix multiplied by $1/r_{12}$ integrated with respect to r.

Differences for the restricted vs the unrestricted method: For the unrestricted method we must use multiple density matrices, this is because it is easier to separate and stop an interaction term being created between an α and β spin set which does not exist. This is simply achieved by maintaining separation for each spin, multiple density matrices and Fock equations. Furthermore when we approach higher number of electron elements such as Boron we need to consider the dipole term in the multipole expansion so we have another function for the r_{12} value along side another density matrix for the P spin state. The second value of r_{12} also needs an accompanying K function for itself and the factor $1/3$ mentioned in section 1.5.

3.2 The Self consistent field loop

With the functions and initial conditions set up it is possible build the main self consistent field (SCF) loop which we will solve iteratively until it reaches a certain error of the previous result - convergence. This is the process of minimising the result as in Eq.11. The loop created has a maximum number of iterations inorder to stop the code in case the wavefunctions do not converge, and inside the loop exists an if statement, this checks previous values of the wavefunction against its previous result to see if we have reached convergence. The next sections will cover the code and logic for each individual atomic system.

Beryllium: For the case of the restricted method with Beryllium it is assumed both electrons in the 1s and the 2s to have the same wavefunction. We then have an initial guess for both wavefunctions which first must be normalized, these are used in our P density matrix function each iteration to calculate a total density matrix. This can then be used to calculate the J and K matrices and we are then able to build our Fock matrix, in this instance it is a single Fock matrix which is solved for the lowest value 1s and second 2s. The program then does all the heavy lifting for us with:

$$\begin{aligned}\text{Eigenvalues, Eigenvectors} =& \\ &\text{eigsh}(\text{Fock}, k = 2, \text{which} =' \text{SA}')\end{aligned} \quad (29)$$

This finds the 'smallest algebriac' eigenvalues and eigenvectors which correspond to our new wavefunctions and the energy levels of the orbitals. Then we simply check our convergence and if the limit has been reached the program can then calculate the total energy of the atom from the individual electron energy. As mentioned earlier this is not simply the sum of all of the individual electron energies as there will be aspects of the problem which will be doubly counted. The program sums the individual orbitals of each electron and adds the core energy of each electron then multiplied by 0.5.

Lithium: The unrestricted method allows us to have different energies for the electrons in each shell which is a more accurate approximation as the electrons on the same orbital behave differently. This is done by having two density matrices, one for each spin value, P_α and P_β. We make these from our three initial guesses, one for each electron. The first consists as you would expect of the $1s\alpha + 2s\alpha$ and the second from the $1s\beta$. These will be used similarly to Beryllium to calculate the J and K values, and in-turn to create the Two Fock matrices that we will solve as seen in Eq.27 & Eq.28. The J matrix uses a total density matrix because it is using just the diagonal terms this is acceptable so it uses P $= P_\alpha + P_\beta$, and calculates as Lithium does - first calculating the charge density ρ and we have one value of J coulomb interaction as per Eq.17. The difference comes from calculating two different exchange K matrices, which calculates one from each spin set to ensure we calculate the exchange between electrons with the same spin. There still needs to be this calculation for the $1s\beta$ electron as the K orbital removes the doubly counted self exchange term in J. Now it is possible to build the two Fock matrices and solve them separately as we did for the single in Beryllium. This again produces the new wavefunction where we continue the process until convergence, giving us the final wavefunctions and energies of our electrons.

Boron: Boron also uses the unrestricted method however the code becomes more complicated due to the dipole shape of the **p**-orbital. The same principle is used, in this case we use a third density matrix for the **p**-orbital with spin α, this will become clear shortly. The same way we totaled the density matrices for the J matrix calculation in Lithium we can repeat the process here for Boron. This time we have to calculate 3, K matrices for 3, Fock equations - this is due to the fact that our l = 1 so we have the centrifugal term for the new orbitals seen in Eq.3. We calculate the different K matrices dependent on what the electrons can 'see'. Simply for the β electrons have the interaction between the two s orbitals which uses the monopole term and P_β density matrix. The α electrons in S orbitals feel two interactions, the interactions between 1s and 2s with the monopole term and the interaction of them both with the Pα orbital, this we have to use the dipole like term in the calculation with the **p** orbital density matrix. Then finally the **p** electron will feel an interaction potential from the 1s and 2s electrons calculated with the P_α density matrix and the dipole interaction term, and not to forget to remove the self interaction. Finally we can construct the 3 Fock equations:

$$F_{sa} = [T + V_{nuc} + J_{mat} - K_{mat_{sa}}] \tag{30}$$

$$F_{sb} = [T + V_{nuc} + J_{mat} - K_{mat_{sb}}] \tag{31}$$

$$F_{sp} = [T + V_{nuc} + V_{centrifugal} + J_{mat} - K_{mat_{sp}}] \tag{32}$$

Then the process is the same as above, solving for the final wavefunctions and energies, except for the final energy calculation, it is essential to include the centrifugal term in the core energy Hamiltonian calculation.

Carbon: The extension then to Carbon was actually relatively simple, as we are working with another electron with α spin, due to Hund's rule of shell filling, most things remain untouched in the code. The only variables to change was the electron density which was now two electrons not one and the interaction term between dipole and spherical for the angular component was 2/3 not 1/3 dependence. Then for the calculation of the energy levels we just have to take into consideration that there are 2, 2p electrons and due to the way the code is written these have to have the same energy level as there is only one solution from the Fock equation.

A note about the cancellation term we have mentioned twice using Lithium as our example. When we construct the coulomb term it is an average field of the three electrons and then we see how each electron interacts with the field, thus creates an exchange between an electron and itself. Through the K matrix we cancel out this unnecessary term.

*See the appendix for an example code.

4 Results and discussion

Having established the required theoretical framework and the numerical method we can now evaluate the results and performance of the code. This section will first discuss the different options for initial conditions and the convergence of these, and then discuss the results of the atomic energy levels.

4.1 Optimisation of initial conditions.

To determine the most effective conditions of the radial grid and efficient initial wavefunctions to use for the project, the test case of Lithium was used. The literature value for the Hartree Fock energy of Lithium is $= -7.4327E_h$. We found that a radial grid defined with $N = 750$ points and a maximum radius of $12.5a.u.$ was sufficient to achieve results close to this value. The relationship between grid density and accuracy is non-linear. While increasing the point density, e.g to $N = 800$, only marginally improves the resolution of the region near the nucleus we are most interested in so the result only improves slightly. The choice of N=750 captures the 1s electrons and the tail of the 2s orbital with sufficient detail and provides a balance between numerical precision and computational efficiency. The program requires initial guesses for the wavefunctions. The different guesses were, the actual grid itself, the r value at each point and, the reduced radial wave functions. For Lithium these are:

$$u_{1s}(r) = 2Z^{3/2} r e^{-Zr} \tag{33}$$

$$u_{2s}(r) = \left(\frac{Z}{2}\right)^{3/2} r \left(1 - \frac{Zr}{2}\right) e^{-Zr/2} \tag{34}$$

See table [1], that both initial conditions converged to the same numerical value of $-7.427E_h$, however the rate at which they did differ.

Initial Guess Type	**Grid Params**	**Description**	**Iterations**	E_{HF} (E_h)
Linear Grid (r)	Li ($Z = 3$)	Linear grid	22	-7.4270
Reduced ($r \cdot \psi_{hyd}$)	Li ($Z = 3$)	Matched shape	20	-7.4270

Table 1: Comparison of convergence rates for various initial trial functions using Lithium as a test case. Literature energy for Lithium: -7.4327 E_h.

Atom	**Elec config**	$1s\uparrow$	$1s\downarrow$	$2s\uparrow$	$2s\downarrow$	$2p_x$	$2p_y$	$2p_z$	E_{tot}	E_{lit}	Δ_{lit}	E_{exp}
Li	$1s^22s^1$	-2.482	-2.464	-0.195	–	–	–	–	-7.4256	-7.4332	-0.0076	-7.4786
Be (Rhf)	$1s^22s^2$	-4.699	-4.699	-0.306	-0.306	–	–	–	-14.5208	-14.5752	-0.0544	-14.6693
Be (Uhf)	$1s^22s^2$	-4.699	-4.699	-0.306	-0.306	–	–	–	-14.5208	-14.5752	-0.0544	-14.6693
B	$1s^22s^22p^1$	-7.6733	-7.6584	-0.5412	-0.4434	-0.3100	–	–	-24.4803	-24.5351	-0.0551	-24.6591
C	$1s^22s^22p^2$	-11.066	-10.909	-1.004	-0.453	-1.135	-1.135	–	-38.4093	-37.7022	0.7071	-37.8574

Table 2: Spin-resolved orbital and total energies compared with literature values. The values are from Hartree-Fock Values Of Energies [10]

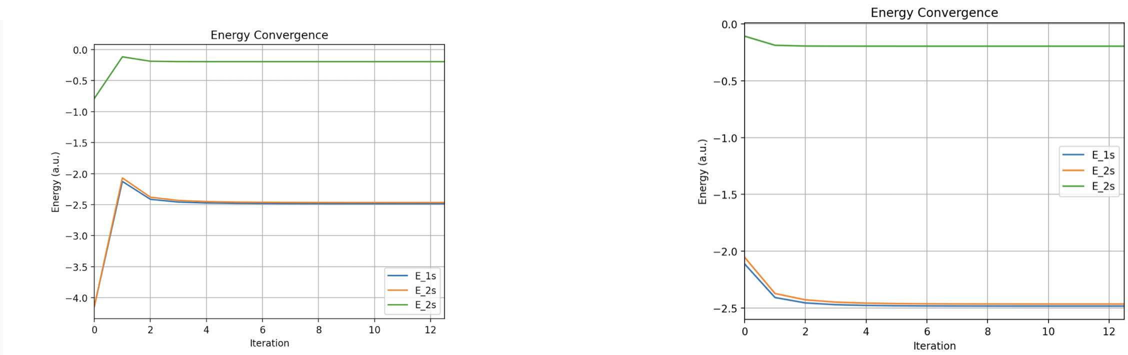
Whilst using the initial conditions of the reduced radial wavefunctions we can see that the convergence rate is faster, however the difference is minimal which is interesting and shows the 'strength' of the underlying physical properties. It is expected that the reduced radial wave functions will converge faster as they have a built in similarity in shape. Therefore, the radial wavefunctions were used for the subsequent atoms in the report to ensure that as the complexity of the problem increases there is an initial backbone present. It is interesting that the convergence rate only differs by 2 iterations.

4.2 Energy levels

The results presented in table [2] demonstrate that the numerical model is consistent. While the accuracy of the total Hartree-Fock energy is limited by the radial grid density, a resolution of N=750 points over the defined radial distance provides a sufficient balance between computational cost and physical detail.

A key validation of the code is that it demonstrates the pauli exclusion principle, the difference between the energy of electrons on the same orbit but having different spin have different bound energy. For instance in the Lithium system the 1s electron with up spin is more tightly bound (more negative) than its 1s down pair electron. This energy gap is a direct consequence of the exchange term in the code. By reducing the energy of parallel spin pairs the model is successful in demonstrating the exchange principle. Similar effects can be seen from the results of the other atoms. The electrons in Carbon follow the trend with a scaling factor, larger gaps appear as there is a larger ratio of electrons with spin up compared with down, the effects seen are greater.

One would expect as increasing the distance between the electrons and nucleus they become less bound, which is the case. It is known that the **p** electrons should orbit the nucleus at a



(a)

(b)

Figure 1: Comparison between the convergence of a) r, and b) Hydrogenic functions.

greater distance than the **s** electrons of the same orbital group so it is strange that the **p** electrons have a larger energy in Carbon and as we will see later in the radial plot these are found on average closer to the nucleus. This could be due to the fact that the code follows Hund's rule and the exclusion principle is so large here that it changes the position of the electrons drastically. However I don't know why they are doing this.

A comparison between the restricted and unrestricted HF method was done for Beryllium as seen in Figure.[2], the values are the same for the restricted vs the un-restricted method, we created the code for both to check whether there would be some evolution and difference in the final energy levels of the electrons if they had there own function. A slight variation in the initial conditions was included. The code would technically produce the same problem but split in two for the system. Szabo and Ostlund [1] did state that the problem may fall back into the same values as the restricted method. Even with the variation of initial wavefunctions, it is not a surprise that there is no difference due to the symmetrical equivalence and .

It is clear from the results that the model is working however accuracy can always be improved, one method to do this would be to implement a logarithmic grid as our basis instead of the equally spaced dr grid currently used.

The radial probability density can also be extracted from our results, defined as $P(r) = |U(r)|^2$, it provides a visual representation of electron position as a distance from the nucleus. Comparing Lithium (Z=3) and Boron (Z=5) highlights the effect of increasing nuclear potential on atomic structure.

As seen in the Figure.[2] the electrons in Boron are much more tightly bound to the nucleus due to the stronger coulomb attraction. The 1s electrons in Boron are twice as close than that of Lithium. The initial small peak of the 2s electrons is also larger in the Boron atom, this is because ?

The effects of the Pauli exclusion principle can also be seen in the radial distribution. Looking at Boron the 2s electrons have a slight splitting and it can be seen that the up spin electron is experiencing a shift closer to the nucleus whereas the down spin electron does not feel the 'push' from the P orbital.

To further validate the accuracy of the model we can apply Koopman's Theorem. Mentioned previously in the theory section, it approximates the first ionisation energy for an atom as the energy level of the individual orbit by assuming the other orbitals do not 'relax'. The results indicate that Koopman's Theorem is a reliable approximation for simpler systems like Lithium, Beryllium and Boron. However, as the number of electrons increases like in Carbon, the approximation becomes less accurate, this could be due to the complex nature of the exchange term and electron averaging when considering interactions between a spherical and dipole shape.

5 Further development

The success of the current model for lower values of Z for producing the ground state energies of atoms and radial distributions suggests that the model could be extended to larger atomic systems, there are several refinements and extensions that would need to be considered to keep the model accurate as we can already see a slight reduction with the model for Carbon due to the complex nature of the interaction. The code should follow the same procedure and increase the K matrix calculations, when we progress to orbitals beyond the **p** shape a different multipole expansion term will have to be calculated and used in the exchange term, for instance the D orbitals will require a quadrupole term. This will increase the complexity of the K matrix calculation for the potential each electron 'sees'. It is essential we still follow Hund's rule of electron filling, starting with parallel spin states before the anti-parallel orbitals as this gives the lowest available energies.

In theory we can keep using the splitting of electron model but as we increase the size of the atom I am not sure the validity of the results will remain the same as we do begin to lose accuracy in the total Hartree energy for Carbon. As we increase the size of the atom it is crucial that the relativistic effects of the atom are also considered. When increasing the size of the atom the relativistic effects become unavoidable, as seen in our lectures hydrogen even has slight relativistic behaviour. When approximating for smaller atoms we will be close to the actual result as seem in .. source with values.. it is only becoming noticeable as we approach values such as Z = 20. Then we will need to use the Dirac equation and a different approximation.

The code could be improved further with a logarithmic grid, this in short fills the orbitals more effectively with points where they are more useful, it tightens the spread to surround the nucleus where the main interactions are taking place so is more accurate. However this requires then the orthonormalisation of the wavefunctions which we did not have to consider with the standard grid used.

6 Conclusion

The primary objective of this project and report was to develop a computational method and provide a background to the Hartree Fock method, building on a topic from the PHYS261 course.

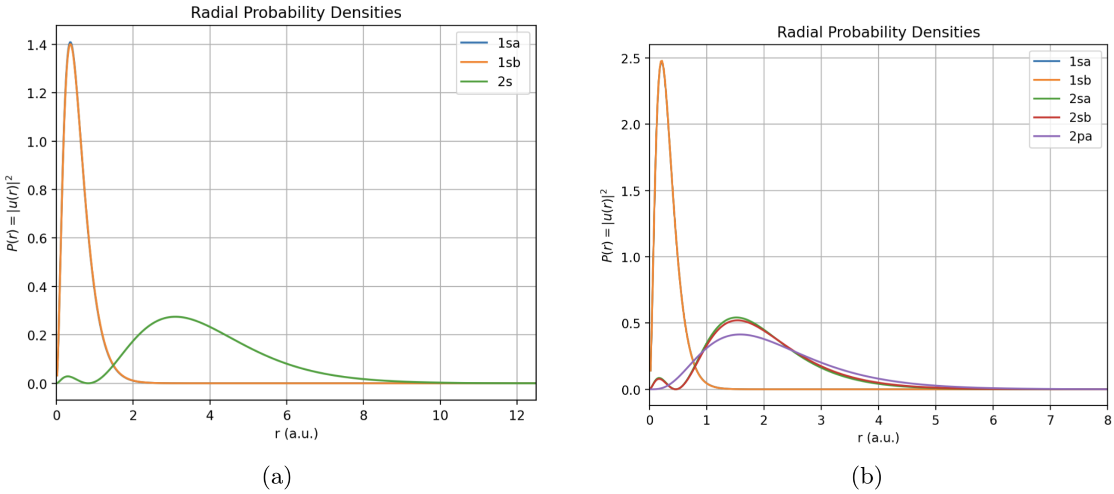


Figure 2: Comparison between the radial probability distributions of a) Lithium, and b) Boron.

The aim was to create a project that would be able to be used as a resource and follow similar methods for coding as that of the course. By using a finite-difference radial grid and a matrix-based approach, the project successfully transitioned from theory to a functional computational tool.

The results from the project are promising, the calculated energy levels and radial distribution agrees with literature and reproduces key physical phenomena.

While the model demonstrates high precision for light atoms, the analysis of heavier systems like Carbon highlights the inherent limits of non-relativistic, mean-field approximations. It is my hope that this report and the accompanying code can serve as a practical resource for future students of PHYS261, these is still a lot more development that can be done with this theory and improvements to the code.

References

[1] 1 Modern Quantum Chemistry - an introduction to advanced electronic structure theory. Available at: Link to Textbook accessed on Feb 1st - 2026.

[2] 2 Central field approximation. Available at: Link to Wiki page.

[3] 3 Hard paper with a radial waveequations Available at: Link to page.

[4] 4 Finite difference method. Available at: Link to page.

[5] 5 Bakermans project Available at: Link to page.

[6] 6 Spin orbital separation Available at: Link to page.

[7] 7 Multipole expansion theory - Prof Morten Førre notes Available at: Link to page.

[8] 8 Spherical Harmonic addition theorum Available at: Link to page.

[9] 9 Basic Spectral Theory Available at: Link to page.

[10] 10 HARTREE-FOCK VALUES OF ENERGIES, INTERACTION CONSTANTS, AND ATOMIC PROPERTIES FOR THE GROUNDSTATES OF THE NEGATIVE IONS, NEUTRAL ATOMS, AND FIRST FOUR POSITIVE IONS FROM HELIUM TO KRYPTON Available at: Link to page.

Background reading sources:

[11] Title - Averaging a dipole field Available at: Link to page.

[12] Title - Spherical harmonic closure relations Available at: Link to page.

[13] Title - The Hartree fock method - Project paper - Time dependence. Available at: Link to page.

[14] Title - Lagrange multipliers Available at: Link to page.

[15] Hartree fock - Back ground Available at: Link to page.

[16] Full method - Extra reading Available at: Link to page.

[17] Another good project - Similar style more advanced Available at: Link to page.

[18] Method and theory stuff Available at: Link to page.

[19] Paper to read Available at: Link to page.

AI

How did I use A.i to aid with the project?

- Initial source searches are quite easy to pinpoint with A.i so I obtained a couple of initial references with prompts such as 'Hartree Fock theory', or 'Hartree Fock code'. Which then have there own list of a lot of sources.

- Code: small fixes and errors are easily spotted, improved code for running speed when trying to achieve better accuracy the run times were long so I ask it to alter a couple of the main lagging components for instance the J and K matrix integration, and some if statements A.i. built quicker components for these to improve run times. Within the multi-pole expansion I had the formula from Mortens notes for the 261 course but did not know how to integrate over the angular component, Gemini provided a target value of 1/3 the integral was over spherical harmonics which I could then calculate the analytical solution for and implement into my code knowing the answer being 1/3 and this drastically improved the Boron value. K matrix building - I could not see how we were able to drop the other wave function from the calculation of the K matrix, it was explained to me by ai that as we are using linear algebra a simple matrix with out the term could be used and when we diaganolise with the guesses it would correct itself through the iterations of the SCF. The import of the code to the appendix into LaTeX format.

- Report : Full spell and grammar check(at end) was done and slight improvement of writing but tried to keep it as most of my own wording.

Appendix: Hartree-Fock Implementation for Lithium

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.sparse import diags
from scipy.sparse.linalg import eigsh

# --- Parameters ---
Z = 3
N = 600
R = 12.5
dr = R / (N + 1)
r = dr * np.arange(1, N + 1)

# --- Functions ---
def build_T_operator(N, dr):
    main = np.full(N, -2.0)
    off = np.ones(N - 1)
    D2 = diags([off, main, off], offsets=[-1, 0, 1], shape=(N, N)) / (dr**2)
    return (-0.5) * D2

def normalize_u(u, r):
    norm = np.sqrt(np.trapezoid(u**2, r))
    return u / norm

def build_r12_matrix(r):
    return 1.0 / np.maximum.outer(r, r)

def build_p_a_matrix(u1, u2):
    return np.outer(u1, u1) + np.outer(u2, u2)

def build_p_b_matrix(u1):
    return np.outer(u1, u1)  # One s down electron

def calculate_J(P, r12_kernel, dr):
    rho = np.diag(P)
    J_vec = np.dot(r12_kernel, rho) * dr
    return diags(J_vec, 0)

def calculate_K_UHF(P, r12_kernel, dr):
    return P * r12_kernel * dr

# --- SCF Setup ---
T = build_T_operator(N, dr)
V_nuc = diags(-Z / r, 0)
kernel = build_r12_matrix(r)

# Initial Guess (Hydrogenic)
u1sa = normalize_u(r*(Z)**1.5 * 2 * np.exp(-Z * r), r)
u1sb = normalize_u(r*(Z)**1.5 * 2 * np.exp(-Z * r), r)
u2sa = normalize_u(r*(Z/2)**1.5 * (1 - Z*r/2) * np.exp(-Z*r/2), r)

MAX_ITER = 50
TOLERANCE = 1e-7
E_1sa_history, E_1sb_history, E_2sa_history = ([] for _ in range(3))

# --- Main SCF Loop ---
for i in range(MAX_ITER):
    Pa = build_p_a_matrix(u1sa, u2sa)
    Pb = build_p_b_matrix(u1sb)
    P = Pa + Pb

    J_mat = calculate_J(P, kernel, dr)
    K_mat_a = calculate_K_UHF(Pa, kernel, dr)
    K_mat_b = calculate_K_UHF(Pb,
```

```
kernel, dr)

    Fock_a = T + V_nuc + J_mat - K_mat_a
    eigenvalues_a, eigenvectors_a = eigsh(Fock_a, k=2, which='SA')

    Fock_b = T + V_nuc + J_mat - K_mat_b
    eigenvalues_b, eigenvectors_b = eigsh(Fock_b, k=1, which='SA')

    new_u1sa = normalize_u(eigenvectors_a[:, 0], r)
    new_u2sa = normalize_u(eigenvectors_a[:, 1], r)
    new_u1sb = normalize_u(eigenvectors_b[:, 0], r)

    e1sa, e2sa = eigenvalues_a[0], eigenvalues_a[1]
    e1sb = eigenvalues_b[0]

    delta_e = abs(e1sa - E_1sa_history[-1]) if E_1sa_history else 999

    E_1sa_history.append(e1sa)
    E_1sb_history.append(e1sb)
    E_2sa_history.append(e2sa)

    if delta_e < TOLERANCE:
        # Total Energy Calculation (0.5 * (E_core + E_orbital))
        H_core = T + V_nuc
        E_core = (np.dot(u1sa, H_core.dot(u1sa)) +
                  np.dot(u1sb, H_core.dot(u1sb)) +
                  np.dot(u2sa, H_core.dot(u2sa))) * dr
        E_total = 0.5 * (E_core + e1sa + e1sb + e2sa)
        break

    u1sa, u1sb, u2sa = new_u1sa, new_u1sb, new_u2sa
```